Robotic-assisted Kasai portoenterostomy for child biliary atresia.

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The Kasai procedure (KPE) is an important treatment for biliary atresia (BA), the most common cause of neonatal obstructive jaundice. To investigate the efficacy of robotic-assisted Kasai portoenterostomy (RAKPE) in patients with BA. Clinical data of 10 patients with BA who underwent RAKPE at the Seventh Medical Center of the People's Liberation Army General Hospital between December 2018 and December 2021 were retrospectively analyzed. One patient underwent Open Kasai portoenterostomy (OKPE) due to intraoperative bleeding. Consequently, nine patients were included in this study. Fifty-two patients who underwent OKPE during the same period served as the control group. Preoperative and postoperative biochemical indexes, surgery-related indexes, and postoperative clearance of jaundice (CJ) were recorded and statistically analyzed. RAKPE was successfully completed in all nine patients, with an average total operative time of 352.2 minutes (including intraoperative cholangiography). Milk feeding resumed on an average 9.89 days postoperatively, and the average time of drainage tube removal was 18.11 days. All patients were followed up for 6 months to 2 years. The liver function indicators and bilirubin levels in 8 patients returned to normal within 3 months after surgery. Three patients developed recurrent cholangitis after discharge, with elevated white blood cell counts, liver function indicators, and bilirubin levels, requiring hospitalization for intravenous antibiotic treatment. The duration of cholangitis ranged from 5 to 8 months post-surgery. To date, no subsequent cases of cholangitis have occurred. All patients have normal liver function and bilirubin levels, with no intrahepatic bile duct dilatation on ultrasonography. Statistical analysis comparing these indicators with those of patients who underwent OKPE showed that the RAKPE group had longer operative times and postoperative drainage tube removal durations. However, there were no significant differences in intraoperative blood loss, postoperative oral milk intake resumption, postoperative hospital stay, or CJ at 3 months post-surgery. RAKPE is technically feasible, safe, and effective for treating BA. Once the technique is mastered, RAKPE may achieve CJ outcomes comparable to those of OKPE.

[Overview of Biliary Atresia].

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Biliary atresia is a progressive, idiopathic, obliterative disease of the extrahepatic biliary tree that presents with biliary obstruction in the neonatal period. It is the most common indication for liver transplantation in children. If untreated, progressive liver cirrhosis leads to death by two years of age. Nowadays, more than 90% of biliary atresia patients survive into adulthood with the development of Kasai portoenterostomy and liver transplantation technology. Early diagnosis is critical since the success rate of the Kasai portoenterostomy decreases with time. This study comprehensively reviews the recent advances in the etiology, classification, prevalence, clinical manifestations, treatment, and prognosis of biliary atresia.

Biliary atresia: how medical complications and therapies impact outcome.

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Biliary atresia (BA) is a progressive fibro-obliterative disease of the extrahepatic biliary tree that presents with biliary obstruction in the neonatal period. Untreated, BA is a uniformly fatal disease and, yet, even with our existing therapies, at least 50% of children with BA will undergo liver transplantation by the age of 2 years. Current treatment strategies are, at best, palliative; they focus on prompt diagnosis, supportive nutritional care and interventions for sequelae. The purpose of this article is to discuss the current treatment paradigm for BA and to assess the impact these strategies have on outcomes. As more children with BA survive into adulthood with their native liver, it is important to understand which factors predict good and poor outcomes.

Biliary atresia.

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Biliary atresia (BA) is a progressive inflammatory fibrosclerosing disease of the biliary system and a major cause of neonatal cholestasis. It affects 1:5,000-20,000 live births, with the highest incidence in Asia. The pathogenesis is still unknown, but emerging research suggests a role for ciliary dysfunction, redox stress and hypoxia. The study of the underlying mechanisms can be conceptualized along the likely prenatal timing of an initial insult and the distinction between the injury and prenatal and postnatal responses to injury. Although still speculative, these emerging concepts, new diagnostic tools and early diagnosis might enable neoadjuvant therapy (possibly aimed at oxidative stress) before a Kasai portoenterostomy (KPE). This is particularly important, as timely KPE restores bile flow in only 50-75% of patients of whom many subsequently develop cholangitis, portal hypertension and progressive fibrosis; 60-75% of patients require liver transplantation by the age of 18 years. Early diagnosis, multidisciplinary management, centralization of surgery and optimized interventions for complications after KPE lead to better survival. Postoperative corticosteroid use has shown benefits, whereas the role of other adjuvant therapies remains to be evaluated. Continued research to better understand disease mechanisms is necessary to develop innovative treatments, including adjuvant therapies targeting the immune response, regenerative medicine approaches and new clinical tests to improve patient outcomes.

Biliary atresia.

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Biliary atresia (BA) is a condition unique to infancy. It results from inflammatory destruction of the intrahepatic and extrahepatic bile ducts. It is the most frequent surgically correctable liver disorder in infancy and the most frequent indication for liver transplantation in paediatric age. Clinical presentation is in the first few weeks of life with conjugated hyperbilirubinaemia (dark urine and pale stools); other manifestations of liver disease, such as failure to thrive, splenomegaly and ascites, appear only later, when surgery is unlikely to be successful. Hence, all infants with conjugated hyperbilirubinaemia must be urgently referred to specialised centres for appropriate treatment. Success of surgery depends on the age at which it is performed. With corrective surgery, followed, when necessary, by liver transplantation, the overall survival rate is approximately 90%. The cause of BA is unknown, but there is evidence for the involvement of infectious, genetic and immunologic mechanisms, which will be discussed in this review.

Sequential treatment of biliary atresia with Kasai portoenterostomy and liver transplantation: a review.

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Biliary atresia is the most frequent cause of chronic cholestasis in infants. When left untreated, this condition leads to death from liver insufficiency within the first 2 yr of life. The modern therapeutic approach consists of a sequential strategy with Kasai portoenterostomy as a first step and, in case of failure, liver transplantation. After portoenterostomy, no more than 20% to 30% of patients will live jaundice-free into adulthood. Illness in another third will be palliated, and these patients have extended survival, delaying liver transplantation to later childhood (2 to 15 yr). The remaining 30% to 40% will not benefit from the Kasai operation and will die of liver failure in infancy. The annual need of liver transplantation for biliary atresia is one case per million people. This indication represents 35% to 67% of the reported series of pediatric liver transplantation and between 5% and 10% of the indications for liver transplantation, all ages included. Approximately four of five children transplanted for biliary atresia will become long-term survivors with good physical and mental development; recurrence of the disease after transplantation has not been observed. Because most candidates are young children (< 3 yr) of small size (< 10 kg), there is a shortage of size-matched donors (which has been alleviated by the use of innovative techniques such as reduced and split livers). The resulting redistribution of the adult donor liver pool is ethically justified by the equal quality of the results after transplantation of a full-size or partial graft.

Improving Poor Outcomes of Children With Biliary Atresia in South Africa by Early Referral to Centralized Units.

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Biliary atresia (BA) is a progressive fibrosing cholangiopathy of infancy, the most common cause of cholestatic jaundice in infants and the top indication for liver transplantation in children. Kasai portoenterostomy (KPE) when successful may delay the requirement for liver transplantation, which in the majority offers the only cure. Good outcomes demand early surgical intervention, appropriate management of liver cirrhosis, and in most cases, liver transplantation. These parameters were audited of children with BA treated at the Steve Biko Academic Hospital (SBAH) in Pretoria, South Africa. All children with BA who were managed at SBAH between June 2007 and July 2018 were included. Parameters measured centered on patient demographics, timing of referral and surgical intervention, immediate and long-term outcomes of surgery, and follow-up management. Of 104 children treated, 94 (90%) were KPE naive. Only 23/86 (26%) of children were referred before 60 days of life and 42/86 (49%) after 120 days. Median time to surgical assessment and surgery was 4 (IQR 1-70) and 5 (IQR 1-27) days post presentation, respectively. The median age at KPE was 91 days (IQR 28-165), with only 4/41 (12%) of KPEs performed before 60 days of life. Of those with recorded outcomes, 12/33 (36%) achieved resolution of jaundice. Only a third of the cohort were referred for transplantation. Children with BA have poor outcomes in the public health sector in South Africa. Late referrals, delayed diagnostics, advanced age at KPE with low drainage rates, poor follow-up, and low transplant rates account for low survival. Early referral to units offering expert intervention at all stages of care, including transplantation, would offer the best outcomes.

Living donor liver transplantation for biliary atresia: An analysis of 2085 cases in the registry of the Japanese Liver Transplantation Society.

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Biliary atresia (BA) is the most common indication for liver transplantation (LT) in pediatric population. This study analyzed the comprehensive factors that might influence the outcomes of patients with BA who undergo living donor LT by evaluating the largest cohort with the longest follow-up in the world. Between November 1989 and December 2015, 2,085 BA patients underwent LDLT in Japan. There were 763 male and 1,322 female recipients with a mean age of 5.9 years and body weight of 18.6 kg. The 1-, 5-, 10-, 15-, and 20-year graft survival rates for the BA patients undergoing LDLT were 90.5%, 90.4%, 84.6%, 82.0%, and 79.9%, respectively. The donor body mass index, ABO incompatibility, graft type, recipient age, center experience, and transplant era were found to be significant predictors of the overall graft survival. Adolescent age (12 to <18 years) was associated with a significantly worse long-term graft survival rate than younger or older ages. We conclude that LDLT for BA is a safe and effective treatment modality that does not compromise living donors. The optimum timing for LT is crucial for a successful outcome, and early referral to transplantation center can improve the short-term outcomes of LT for BA. Further investigation of the major cause of death in liver transplanted recipients with BA in the long-term is essential, especially among adolescents.

Biliary atresia and survival into adulthood without transplantation: a collaborative multicentre clinic review.

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Biliary atresia is a progressive biliary injury which occurs only in infants. To review the experience of patients surviving into adulthood without the need for liver transplantation in childhood. A multicentre review of patients with biliary atresia treated surgically who survived into adulthood without the need for transplantation. Twenty-two patients were identified across four centres. Median age at the last follow-up was 25 years (range: 18-46), and 21 patients had clinical features of portal hypertension. At last follow-up values of liver enzymes varied from normal to 15 × the upper limit of normal (ULN) for ALT (median 2.11 × ULN) and 9 × the ULN for ALP (median 2.02 × ULN). Six patients had a serum bilirubin > 50 µmol/l. Pruritus and jaundice were noted in 8 of 20 patients (40%) and 11 of 22 patients (50%) respectively. Thirteen patients (59.1%) were shown to have imaging features of sclerosing cholangitis, with strictures of intrahepatic bile duct(s) (IHBD), dilatation of IHBD (n = 8), or stone(s) within the IHBD (n = 5). A history of presumed bacterial cholangitis was present in 11 patients (50%). Successful pregnancies were recorded in three of fourteen female patients. Four patients underwent transplant between the ages of 20-27 years. Twenty-one patients (95.5%) were alive, including 18 (81.8%) with their native liver at the time of last follow-up. Some patients treated for biliary atresia will survive into adulthood with their native liver, but commonly with secondary biliary disease including cholangitis and portal hypertension.

Biliary atresia and orthotopic liver transplantation. 11 years of experience in Geneva.

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Biliary atresia (BA) is a congenital malformation or an evolutive inflammatory process which, without treatment, leads to cirrhosis, hepatic failure and death within two years of birth. The literature gives a survival rate of 60% at five years and 25% to adulthood after an initial operation performed for BA. 30% of children do not survive beyond two years of age. BA has become the most frequent indication for liver transplantation (LT) in children. With LT, survival expectancy is 90%. Results of the operation designed for BA remain unsatisfactory, and seem to depend on the age of the infants, as well as on other factors such as liver histology, and centre experience. Since 1989, onset of the paediatric hepatic transplantation program in Geneva, to July 2000, 20 children have been referred for initial treatment of BA, and 26 for possible hepatic transplantation after initial treatment done in another centre. The aim of the current study is to analyse our own results of the initial operation and to present the results of liver transplantation in this particular group of patients. All the patients with a BA are included in this study. The initial operation for BA yielded 43% favourable outcome at five years and the survival in this group following LT reached 91.3% survival. The importance of the age of the patient at time of initial operation is underlined.